

Seongho Kim

Curriculum Vitae

CONTACT INFORMATION

School of Electrical and Electronic Engineering
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EDUCATION

Yonsei University, Seoul, Republic of Korea

Ph.D. Candidate in Electrical and Electronic Engineering, March 2025 to Present

Compiler Research Laboratory (CoreLab), Advisor: Prof. Hanjun Kim

GPA: 4.27 / 4.3, Full scholarship

Yonsei University, Seoul, Republic of Korea

Bachelor of Engineering in Electrical and Electronic Engineering, March 2019 to February 2025

Yonsei University Semiconductor Specialist Track

GPA: 3.72 / 4.3

EXPERIENCE

Research Assistant, March 2025 to Present

Compiler Research Laboratory (CoreLab), Advisor: Prof. Hanjun Kim, Yonsei University

- Developed Recifhe, a multi-level compiler for fully homomorphic encryption (FHE) that optimizes programs at both the ciphertext and polynomial levels (arXiv 2026, first author)
- Built the end-to-end pipeline of HECATE, an open-source FHE compiler: Python-level workload tracing, MLIR-based transformation, and execution on CPU, GPU, and custom accelerators
- Implemented MLIR compiler passes that automate GPU kernel optimization for FHE workloads
- Designed a sparsity-aware matrix-vector multiplication algorithm for RNS-CKKS that outperforms state-of-the-art implementations
- Designed a tensor abstraction that exposes ciphertext layout to the compiler for explicit layout optimization (CGO 2026 SRC)
- Co-developed CoV, a compiler-runtime verification chain for LLM-based vectorization in LLVM that vectorizes 13.9% more loops than LLVM -O3 (CGO 2026)

Undergraduate Research Assistant, June 2024 to February 2025

Compiler Research Laboratory (CoreLab), Advisor: Prof. Hanjun Kim, Yonsei University

- Profiled kernel-level execution patterns of privacy-preserving machine learning workloads to identify GPU bottlenecks

Undergraduate Research Lead, March 2024 to June 2024

Design and Architecture for Intelligent Systems Laboratory, Advisor: Prof. Joon-Sung Yang, Yonsei University

- Led GPU kernel profiling across LLM architectures to identify operation-level performance bottlenecks (IEEE OJCS 2025, first author)

Undergraduate Research Intern, December 2023 to February 2024

Scalable Computer Architecture Laboratory, Advisor: Prof. Jung Ho Ahn, Seoul National University

- Performed architectural and kernel-level analysis of Llama 2 7B and representative deep learning models

Undergraduate Research Intern, January 2023 to June 2023

Circuit and System Laboratory, Advisor: Prof. Tae Wook Kim, Yonsei University

- Designed a pulse generator for an ultra-wideband (UWB) transmitter

Information Systems Officer (Mandatory Military Service), October 2020 to July 2022

Headquarters, Republic of Korea Air Force

PROJECTS

Maintainer, HECATE Compiler (Open Source), 2025 to Present

Compiler Research Laboratory (CoreLab), Yonsei University

github.com/corelab-src/hecate-compiler

- Maintain HECATE, an open-source MLIR-based compiler for privacy-preserving machine learning on RNS-CKKS FHE: Python frontend, compiler transformations, and CPU, GPU, and accelerator backends

RECOGNITION

- Emerging Engineering Researcher, Institute for the Promotion of Engineering and Science in Korea (IPESK), September 2026
- Bronze Prize, 32nd Samsung Humantech Paper Award, Samsung Electronics, February 2026 (with Hyunho Kwon, Ju Min Lee, and Hoyun Youm)
- Emerging Engineer, IPESK, December 2024

ACTIVITIES

PROFESSIONAL SERVICE

- Session Volunteer, 58th IEEE/ACM International Symposium on Microarchitecture (MICRO 2025), Seoul, Republic of Korea, October 2025.

PUBLICATIONS

PREPRINTS AND TECHNICAL REPORTS

- [1] Seongho Kim, Heelim Choi, Jaemin Kim, Seonyoung Cheon, Dongkwan Kim, Jaeho Lee, Hoyun Youm, Dongyoon Lee, Hanjun Kim, and Yongwoo Lee, “Ciphertext- and Polynomial-Level Optimization for Fully Homomorphic Encryption,” *arXiv preprint arXiv:2607.15750*, July 2026.

REFEREED JOURNAL PUBLICATIONS

- [2] Seongho Kim, Jihyun Moon, Juntaek Oh, Insu Choi, and Joon-Sung Yang, “Survey and Evaluation of Converging Architecture in LLMs Based on Footsteps of Operations,” in *IEEE Open Journal of the Computer Society (OJCS)*, July 2025.
[IF 8.2, Q1, JCR 2024]

REFEREED CONFERENCE PUBLICATIONS

- [3] Hyunho Kwon, Sanggyu Shin, Ju Min Lee, Hoyun Youm, Seungbin Song, Seongho Kim, Hanwoong Jung, Seungwon Lee, and Hanjun Kim, “Compiler-Runtime Co-operative Chain of Verification for LLM-Based Code Optimization,” in *Proceedings of the 2026 International Symposium on Code Generation and Optimization (CGO)*, January 2026.

REFEREED POSTER PUBLICATIONS

- [4] Seongho Kim and Hanjun Kim, “Tensor Abstraction Enabling Explicit Layout Optimization in Homomorphic Encryption,” in *Student Research Competition (SRC), the 2026 IEEE/ACM International Symposium on Code Generation and Optimization (CGO)*, February 2026.

PATENTS

- [5] Hanjun Kim, Hoyun Youm, Seonyoung Cheon, Dongkwan Kim, Sungwoo Yun, Chan Lee, Seongho Kim, and Yongwoo Lee, “Electronic Apparatus for Controlling Placement of Bootstrapping Operations in Fully Homomorphic Encryption Based on Range Analysis, and Method Thereof,” KR Patent App. 10-2026-0024507, February 2026.